

SET-I

1. Data Structure suitable for an application is discussed in:
A. Data design
B. Architectural design
C. Procedural design
D. Interface design
2. Which of the followings is a tool in design Phase?
A. Abstraction
B. Refinement
C. Information hiding
D. All of the above
3. In object-oriented design of software, which of the following is NOT true?
A. Objects inherit the properties of the class
B. Classes are defined based on the attributes of the object
C. Objects can belong to two classes
D. Classes are always different from each other
4. Design Phase includes
A. Data, architectural and procedural design
B. Architectural, procedural and interface design only
C. Data, architectural and interface design only
D. Data, architectural interface and procedural design
5. The organized process or set of steps that needs to be followed to develop an information system is known as:
A. Analytical cycle
B. Design cycle
C. Program specification
D. System development Life cycle
6. Translating the problem statement into a series of sequential steps describing what the program must do is known as:
A. Coding
B. Debugging
C. Creating the algorithm
D. Writing documentation
7. The first step in system development life cycle (SDLC) is:
A. Analysis
B. Design
C. Problem/opportunity identification
D. Development and documentation
8. In a multiuser database, if two users wish to update the same record at the same time, they are prevented from doing so by
A. Jamming B. Password
C. Documentation D. Record lock
9. Which of the following contains complete record of all activities that affected the contents of a database during a certain period of time?
A. Transaction log B. Query Language
C. Report writer D. DML
10. Relations produced from E-R model will always be in
A. 1st normal form B. 2nd normal form
C. 3rd normal form D. 4th normal form
11. Who of the following is more concerned about the conceptual level of the DBMS?
A. DBA
B. END user
C. System programmer
D. Client

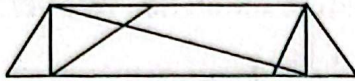
12. Consider the following set of functional dependencies of the schema (A, B, C), $A \rightarrow BC$, $B \rightarrow C$, $A \rightarrow B$, $AB \rightarrow C$. The canonical cover for this set is
 - A. $A \rightarrow BC$ and $B \rightarrow C$
 - B. $A \rightarrow BC$ and $AB \rightarrow C$
 - C. $A \rightarrow BC$ and $A \rightarrow B$
 - D. $A \rightarrow B$ and $B \rightarrow C$
13. What is the name of operating system that reads and reacts in terms of actual time?
 - A. Batch system
 - B. Quick response system
 - C. Real time system
 - D. Time Sharing system
14. To access the services of operating system, the interface is provided by the
 - A. System calls
 - B. API
 - C. Library
 - D. Assembly instruction
15. Suppose that a process is in "Blocked" state waiting for some I/O service. When the service is completed, it goes to the
 - A. Running state
 - B. Ready State
 - C. Suspended state
 - D. Terminated state
16. Memory management technique in which system stores and retrieves data from secondary storage for use main memory is called?
 - A. Fragmentation
 - B. Paging
 - C. Mapping
 - D. Scheduling
17. Which of the following conditions is required for deadlock to be possible?
 - A. Mutual exclusion
 - B. A process may hold allocated resources while awaiting assignment of other resources
 - C. No resource can be forcibly removed from a process holding it
 - D. All of the mentioned
18. When an attempt is to make a machine or network resource unavailable to its intended users, the attack is called?
 - A. Denial-of-service attack
 - B. Slow read attack
 - C. Spoofed attack
 - D. Starvation attack
19. In distributed systems, link and site failure is detected by
 - A. Polling
 - B. Handshaking
 - C. Token passing
 - D. None of the mentioned
20. A formal grammar is a for writing strings:
 - A. Set of rules
 - B. Set of functions
 - C. Both A and B
 - D. None of the above
21. Context free language are not closed under:

A. Union	B. Concatenation
C. Closure	D. Iteration
22. Push down machine represents:
 - A. Type 0 grammar
 - B. Type 1 grammar
 - C. Type 2 grammar
 - D. Type 3 grammar

23. Set of regular languages over a given alphabet set is not closed under:
- Union
 - Complementation
 - Intersection
 - None of the above
24. A top-down parser generates
- Right-most derivation
 - Right-most derivation in reverse
 - Left-most derivation
 - Left-most derivation in reverse
25. Which of the following is used for grouping of character into tokens?
- Parser
 - Code optimizer
 - Code generator
 - Scanner
26. Minimum hamming distance method is used for grouping of characters into tokens?
- Syntactic errors
 - Semantic errors
 - Algorithm errors
 - Transcription errors
27. An optimizing compiler
- Is optimized to occupy less space
 - Is optimized to take less time for execution
 - Optimizes the code
 - None of the above
28. Engineering drawing commonly applies
- Oblique projection
 - Orthographic projection
 - Perspective projection
 - None of the above
29. A Circle, if scaled only in one direction becomes a
- Parabola
 - Hyperbola
 - Ellipse
 - Circle
30. A transformation which maps one vector to another and preserves linear combinations is known as
- Linear transformation
 - Non-linear transformation
 - Simple transformation
 - Combination transformation
31. If the flux per pole of a shunt-wound DC generator is increased, and all other variables are kept the same, the speed
- Decrease
 - Stays the same
 - Increases
 - None of above
32. In a series-wound generator running at constant speed, as the load current increases, the terminal voltage
- Increases
 - Decreases
 - Stays the same
 - None of above
33. A sine wave voltage is applied across an inductor. When the frequency of the voltage is decreased, the current:
- Does the change
 - Is decreased
 - Is increased
 - Momentarily goes to zero
34. Three 15 mH inductors are in series. The total inductance is:
- 15 mH
 - 30 mH
 - 45 mH
 - 5 mH
35. What is the Maximum power rating for LEDs
- 150 mW
 - 500 mW
 - 1 w
 - 10 w
36. Which of the following devices can check the condition of a semiconductor diode?
- Digital Display Meter (DDM)
 - Multimeter
 - Curve tracer
 - All of the above

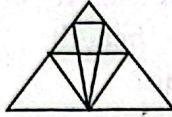
37. In which of the following is the light intensity measured?
 A. Candela B. Efficacy
 C. Flux D. Illumination
38. Which of the following is not the promise of artificial neural network?
 A. It can explain result
 B. It can survive the failure of some nodes
 C. It has inherent parallelism
 D. It can handle noise
39. Image quantized with insufficient brightness levels will lead to the occurrence of?
 A. Pixilation B. Blurring
 C. False contours D. None of above
40. Which of the following is GIS package?
 A. Arc GIS B. Arcinfo
 C. Idrisi 32 D. All of the above
41. When you look at e-commerce and e-business, which is the most realistic relationship between these two concepts?
 A. E-commerce is subset of e-business
 B. E-business is the subset of e-commerce
 C. E-commerce has some overlap with e-business
 D. E-commerce is broadly equivalent to e-business
42. The first in the Monte Carlo simulation process is to
 A. Generate random numbers
 B. Steps cumulative probability distributions
 C. Establish random number intervals
 D. Setup probability distributions
43. Which of the following, statistical methods are commonly used to analyze simulation results?
 A. T-test
 B. Regression analysis
 C. Analysis variance
 D. All of the above
44. Which of the followings is not the correct benefit of distributed computing?
 A. Resource sharing
 B. Performance
 C. Availability
 D. Security
45. In asymmetric key cryptography, the private key is kept by
 A. Sender
 B. Receiver
 C. Sender and receiver
 D. All the connected devices to the network
46. In simulation and modeling..... is a foundation and sometimes intangible notion covering the recognition, observation, nature, and stability of patterns and relationship of entities?
 A. Base
 B. Structure
 C. Concrete
 D. Sample
47. Which may be estimated either in terms of KLOC (Kilo Line of Code. or number of function points in the software?
 A. Time estimation
 B. Effort estimation
 C. Cost estimation
 D. Software size estimation

48. Find the number of triangles in the given figure



- A. 8
B. 10
C. 12
D. 14

49. Find the number of triangles in the given figure



- A. 12
B. 18
C. 22
D. 26

50. Pointing to a man, a woman said "His mother is the only daughter of my mother". How is the woman related to the man?

- A. Mother
B. Daughter
C. Sister
D. Grand Mother

51. Arun said that "This girl is the wife of the grandson of my mother". Who is Arun to the girl?

- A. Grand Mother
B. Husband
C. Father-in-law
D. Father

52. Select the word which can be formed using the letters of the word "COLLABORATION"

- A. LABOUR
B. CAUTION
C. NATION
D. BALLOON

53. If PANCTUAL can be written as 47819765, how can LUNA be written in that code?

- A. 7586
B. 5678
C. 5786
D. 5867

54. A series is given below with one missing number. Choose the correct alternative from the given ones that will complete the series:

138, 142, 146, 150, ?, 142, 138

- A. 148
B. 140
C. 146
D. 144

55. Find the related number to complete the analogy?

122 : 222 : 442 : ?

- A. 552
B. 882
C. 884
D. 886

56. Find the odd one out from the given alternative?

- A. Tortoise
B. Crab
C. Frog
D. Fish

57. Which one of the letters when sequentially placed in the gaps in the given letter series shall complete it?

_ bas _ asb _ s _ a _ ba

- A. sbaba
B. sabbs
C. sbabs
D. bsabs

58. What is the largest integer value of X that satisfies the inequality $24/30 > X/24$?

- A. 8
B. 10
C. 18
D. 19

59. Choose the correct answer for what the two possible values of X are:

solve $|4x-6| = 10$ for x

- A. 4, -1
B. -4, 1
C. 1, -1
D. 0, 4

60. Kathy was twice as old as Jim 2 years ago. Today Jim is j years old. In terms of j, how old was Kathy 2 years ago?

- A. $2(j-2)$
B. $2j-2$
C. $2(j+2)$
D. $j(2+2)$

61. If $7y + 9$ represents an odd integer, which of the followings represents the next smaller odd integer?

- A. $7(y-2)$
B. $7(y+1)$
C. $7(y+2)$
D. $7(y+3)$

62. a, b, c and d are playing cards. a and b are partners. d faces towards north if a face towards west, then who faces towards south?
A. c B. b
C. d D. a
63. Juan walked 3 more miles than Rebecca. Rebecca walked 4 times as far as William, William walked 2 miles. Which of the following mathematical expressions is equivalent to the number miles Juan walked?
A. $3 \times 4 \times 2$ B. $(2+4) \times 3$
C. $4(2) + 3$ D. $4+3+2$
64. For the following question choose the best answer from the options given below: What is $\frac{1}{5}$ of 16% of \$24,00?
A. \$ 160 B. \$ 768
C. \$ 3840 D. \$ 4, 032
65. For the following question choose the best answer given below.
Mark is older than Frank, but younger than David. If m, f and d represent the ages of mark, Frank and David, respectively, which of the following is true?
A. $d < f < m$ B. $f < m < d$
C. $d < m < f$ D. $d < d < m$
66. If $w = xyz$, then which of the following is an expression for z in terms of w, x and y?
A. xy/w B. w/xy
C. wxy D. $w - xy$
67. Look at this series: 7, 10, 8, 11, 9, 12..... what number should come next?
A. 7 B. 10
C. 12 D. 13
68. Shyam walked 2 km west of his house and then turned south covering 4 km. finally, he moved 3 km. towards east and then again 1 km west how far is he from his initial position?
A. 10 km B. 9 km
C. 2 km D. 4 km
69. What is the minimum header size of an IP packet?
A. 16 bytes
B. 10 bytes
C. 20 bytes
D. 32 bytes
70. What does router do in a network?
A. forward a packet to all outgoing links
B. Forward a packet to the next free outgoing link.
C. Determines on which outgoing link a packet is to be forwarded
D. Forwards a packet to all outgoing links except the originated link.
71. What is the full form of RAID?
A. Redundant Array of Independent Disks
B. Redundant Array of Important Disks
C. Random Access of Independent Disks
D. Random Access of Important Disks
72. Each IP packet must contain
A. Only source address
B. Only destination address
C. Source and destination address
D. Source of destination address
73. How many layers are in the TCP/IP model?
A. 4 layers B. 5 layers
C. 6 layers D. 7 layers

74. DHCP is the abbreviation of
- Dynamic Host Control Protocol
 - Dynamic Host Configuration protocol
 - Dynamic Hyper Control Protocol
 - Dynamic Hyper Configuration Protocol
75. A normal C++ operator that may act differently on newly defined data types is called
- Glorified
 - Encapsulated
 - Classified
 - Overloaded
76. The member function can always access the data in
- The object of which it is a member
 - The class of which it is member
 - Any object of the class of which it is a member
 - The public part of its class
77. An enumerated data type brings together a group of
- Integer with user defined names
 - Related data type
 - Items of different data types
 - Constant values.
78. In a linked List
- Each link contains a pointer to the next link
 - An array of pointer points to the links
 - Each link contains data or a pointer to data
 - Link are stored in an array
79. When a language has the capability to produce new data types, it is called
- Reprehensible
 - Encapsulated
 - Extensible
 - None of above
80. The dot operator connects which of the following two entities?
- Class member and class object
 - Class object and a class
 - Class and member of that class
 - Class object and member of that class
81. The First artificial intelligence programming language was
- BASIC
 - IPL (Inductive Logic Programming)
 - LISP
 - FORTRAN
82. Which of the following search belongs to totally ordered plan search?
- Forward state-space search
 - Hill-climbing search
 - Depth-first search
 - Breadth-first search
83. General algorithm applied on game tree for making decision of win/ lose is
- DFS/BFS Search Algorithms
 - Heuristic search algorithms
 - MIN/MAX algorithm
 - Greedy search algorithm
84. Which of the following data structure is linear data structure?
- Trees
 - Graphs
 - Arrays
 - None of above
85. In a Heap tree
- Values in a node is greater than every value in left sub tree and smaller than right sub tree
 - Values in a node is greater than every value in children of it
 - Both of above conditions applies
 - Values in a node is smaller than every value in children of it

86. Stack is also called as
 A. Last in First Out
 B. First in last out
 C. First in first out
 D. Last in last out
87. Which of the following data structure can't store the non-homogeneous data elements?
 A. Arrays B. Records
 C. Pointers D. Stacks
88. To represent hierarchical relationship between elements, which data structures is suitable?
 A. Dequeue
 B. Priority
 C. Tree
 D. Graph
89. is very useful in situation when data have to be stored and then to be retrieved in reverse order?
 A. Stack B. Queue
 C. List D. Linked list
90. Dynamic memory cells are constructed using
 A. Transistor B. Flip-flop
 C. MOSFETS D. FETS
91. Which of the following interrupts has the lowest priority?
 A. RST 5.5 B. RST 7.6
 C. TRAP D. INTR
92. The process of organizing the memory into two banks to allow 8 and 16-bit data operation is called
 A. Bank switching
 B. Indexed mapping
 C. Two-way memory interleaving
 D. Memory segmentation
93. Which of the following statement is incorrect for micro-processor system using I/O mapped I/O?
 A. I/O address space is greater
 B. I/O and memory spaces are distinct
 C. Memory space available
 D. Not all data transfer instructions are available
94. The ALU of computer normally contains a number of high-speed storage elements called
 A. Semiconductor memory
 B. Registers
 C. Hard disk
 D. Magnetic disk
95. In which addressing mode, the effective address of the operand is generated by adding a constant value to the contents of a registers?
 A. Absolute mode
 B. Indirect mode
 C. Immediate mode
 D. Index mode
96. In which of the following gates, the output is 1, if and only if at least one input is 1
 A. NOR B. AND
 C. OR D. NAND
97. The time required for a gate to change its state is called
 A. Propagation time
 B. Rise time
 C. Decay time
 D. Changing time
98. Which table shows the logical state of digital circuit output for every possible combination of logical states in the input?
 A. Function table B. ASCII table
 C. Routing table D. Truth table

99. A demultiplexer is used to

- A. Route the data from single input to one of many output
- B. Perform serial to parallel conversion
- C. Both A and B
- D. Select data from several inputs and route them to single output

100. How many flip-flops are required to make a MOD-32 binary counter?

- A. 5
- B. 3
- C. 6
- D. 45

ANSWER SHEET

1.A	2.D	3.C	4.D	5.D	6.C	7.C	8.D	9.A	10.A
11.A	12.D	13.C	14.A	15.B	16.B	17.D	18.A	19.B	20.A
21.D	22.C	23.D	24.C	25.D	26.A	27.D	28.B	29.C	30.A
31.A	32.A	33.C	34.C	35.A	36.D	37.A	38.A	39.C	40.D
41.A	42.D	43.D	44.D	45.B	46.B	47.D	48.D	49.B	50.A
51.C	52.D	53.C	54.C	55.B	56.D	57.C	58.-D	59.A	60-A
61.B	62.A	63.C	64.-B	65.B	66-B	67.B	68.D	69.C	70.C
71.A	72.C	73.B	74.B	75.D	76.B	77.D	78.A	79.C	80.D
81.C	82.A	83.C	84.C	85.B	86.A	87.A	88.C	89.A	90.C
91.A	92.D	93.A	94.B	95.D	96.C	97.A	98.D	99.A	100.A